

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Annexure A

Total Marks:30

Constraint-Based Problem Solving

Code of Conduct:

I Sree B C 192524023 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B. B.

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability.
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

① Internet Connection, Networking Devices, and Security for a Remote Rural Branch .

For a remote rural branch where wired high speed Internet is unavailable; a 4G/5G business connection or satellite Internet is suitable.

- * Primary connection: 4G/5G fixed wireless Internet, if cellular coverage is available.

- * Backup connection: Satellite Internet or a second cellular provider.

- * Use dual-WAN failover so that if the primary connection fails, traffic automatically switches to the backup.

- * A business grade connection with a suitable data plan should be selected because cloud applications and video conferencing require stable bandwidth.

Remote Rural Branch Office

- * Uses 4G/5G or satellite internet as the main connection
- * Uses 4G/5G if good mobile coverage is available
- * Uses satellite internet if mobile coverage is poor.
- * Use a dual-WAN router for backup connectivity
- * Install a Wi-Fi router/access point for office users.
- * Uses a VPN to securely connect with headquarters
- * Uses a Firewall to block unauthorized traffic.
- * Uses WPA3 for secure wireless access.
- * 4G/5G provides good speed at lower cost.
- * Satellite provides connectivity in remote areas.
- * Backup connectivity improve reliability.
- * The network can be upgraded to fibre in the future.
- * This design balances cost, reliability, security and scalability.

Wireless Network Design for a University with 3000 Students

Introduction

A university with more than 3000 students requires a wireless network capable of supporting a large number of simultaneous users in classrooms, laboratories, libraries, and hostels. Since the number of wireless access points is limited by the budget, the network should use careful access-point placements, appropriate frequency bands, centralized management and efficient authentication.

Access Point Placements

Access points should be placed mainly in classrooms, laboratories and hostels where the number of users is high. Proper placements help to provide maximum coverage with few access points.

Frequency Selection

The 5GHz band should be preferred because it provides higher speed and less interference. The 2.4GHz band can be used where longer coverage is required. WiFi 6 can improve performance for many simultaneous users.

Authentication

WPA2 - Enterprise or WPA3 - Enterprise with 802.1X and RADIUS should be used for secure student and staff authentication. Separate VLANs can be created for students, staff, guests and IoT devices.

Justification

Proper channel selection and transmit power control reduce interference. Centralized management improves performance and security. This design provides good coverage, supports many users, and reduces cost by placing access points according to user density.

Financial Institution

Introduction

A financial institution handles highly sensitive customers and financial information, so its network must provide strong security while maintaining good performance and reasonable operational costs. The recommended architecture should use network segmentation, firewalls, strong authentication, intrusion detection, encryption and continuous monitoring.

Network Segmentation

The network should be divided into separate VLAN's for employees, servers, finance systems, guests and management devices. Sensitive databases should be isolated from normal users.

Firewall

A next generation firewall should be placed between the Internet and Internal network. A DMZ can be used for public services such as web servers.

Authentication

Multi factor authentication, 802.1x, role based access control and VPN encryption should be used to protect sensitive customer information and prevent unauthorized access.

Intrusion Detection

An IDS/IPS should monitor traffic, detect suspicious activities and block threats. Centralized logging can be used to monitor security events.

Justification

This architecture improves cybersecurity, protects sensitive financial data, supports compliance requirements, maintains network performance and controls operational costs.

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